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A Comprehensive Review of Deep Learning Techniques for Anomaly Detection in IoT Networks: Methods, Challenges, and Datasets

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ABSTRACT

With the rapid growth of the Internet of Things (IoT) and the widespread deployment of smart connected devices, ensuring the security of these networks has become a critical challenge. Anomaly detection is considered one of the most effective techniques for identifying abnormal behaviors and cyber-attacks in IoT networks. In recent years, deep learning techniques have gained significant attention in this domain due to their powerful capabilities in automatic feature extraction and modeling complex patterns. This review article provides a comprehensive overview of deep learning methods applied to anomaly detection in IoT networks. Various deep architectures including CNNs, LSTMs, autoencoders, GANs, and hybrid models are analyzed and compared. In addition, commonly used datasets such as CICIDS2017, BoT-IoT, NSL-KDD, and TON_IoT are introduced and evaluated in terms of their quality and suitability for deep learning-based models. Key challenges including the lack of real-world data, high resource consumption, vulnerability to adversarial attacks, and lack of interpretability are also discussed. Finally, potential future research directions are suggested to enhance the performance and real-world applicability of deep learning-based anomaly detection systems in IoT environments.

1 | Introduction

The Internet of Things (IoT) has emerged as a transformative technology ecosystem, connecting billions of devices that collect, transmit, and act on data autonomously. These devices—ranging from wearable health monitors and smart thermostats to industrial actuators and autonomous vehicles—are deeply integrated into daily life and critical infrastructure. According to industry forecasts, the number of internet-connected devices is expected to surpass 41 billion by 2026, signaling an era of hyper-connectivity and data-driven intelligence.

While IoT brings numerous advantages in automation, efficiency, and decision-making, it simultaneously introduces major cybersecurity concerns. The distributed, heterogeneous, and resource-constrained nature of IoT systems makes them especially vulnerable to a broad spectrum of cyber threats, including Distributed denial of service (DDoS), man-in-the-middle (MITM), spoofing attacks, ransomware, and data exfiltration. The risk is amplified by the fact that many IoT devices are deployed with minimal security configurations, often lacking mechanisms for regular updates or robust authentication.

Traditional network security mechanisms—such as firewalls, rule-based intrusion detection systems, and signature-based antivirus tools—are ill-suited for dynamic IoT environments due to their static nature, inability to scale, and dependence on prior knowledge of threats. In this context, anomaly detection plays a

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critical role as a proactive defense strategy by identifying deviations from normal behavior that may signal previously unseen or zero-day attacks. However, classical machine learning-based anomaly detection techniques often rely on handcrafted features, shallow models, and limited scalability, which restricts their applicability in large-scale, real-time IoT scenarios.

Recent advances in deep learning (DL) offer promising alternatives by enabling models to automatically learn meaningful representations from high-dimensional and sequential data, adapt to evolving threats, and generalize across different deployment contexts. Models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders (AEs), generative adversarial networks (GANs), and various hybrid architectures have demonstrated superior performance in capturing complex patterns in IoT traffic and detecting subtle anomalies.

Despite this progress, there is a lack of comprehensive and comparative reviews that systematically assess these models in the specific context of IoT anomaly detection. Many existing studies focus on individual methods or datasets, limiting broader insights into their real-world applicability, limitations, and trade-offs.

This review aims to fill this research gap by:

- Providing a structured overview of state-of-the-art deep learning techniques used for anomaly detection in IoT networks.
- Comparing their architectures, advantages, and limitations.
- Evaluating their performance on widely used benchmark datasets, including CICIDS2017, BoT-IoT, NSL-KDD, and TON_IoT.
- Discussing open challenges such as data scarcity, computational constraints, adversarial robustness, and model interpretability.
- Outlining future research directions for building more robust, efficient, and deployable deep learning-based intrusion detection systems (DL-IDS) tailored to the unique characteristics of IoT environments.

Figure 1 presents a conceptual framework of deep learning-based anomaly detection in IoT networks, illustrating the sequential pipeline from data acquisition and preprocessing to model training, detection, and alert generation. This framework serves as the foundation for the comparative analysis and discussions presented in the subsequent sections of this review.

Figure 1 illustrates a high-level pipeline for anomaly detection in IoT networks, comprising four key stages: IoT networks, datasets, anomaly detection system, and associated challenges and future directions. To enhance clarity, we provide detailed annotations and descriptions for each stage:

- IoT networks: This stage encompasses diverse IoT devices and environments, including smart homes, surveillance cameras, wearables, and industrial sensors. These devices generate heterogeneous and multi-modal traffic data characterized by varying protocols, sampling rates, and noise levels. Assumptions include the presence of both normal and anomalous behaviors, with potential temporal and spatial correlations.
- Datasets: Representative benchmark datasets such as CICIDS2017, BoT-IoT, and NSL-KDD are highlighted. These datasets provide labeled network traffic capturing various attack types and normal operations. Preprocessing steps applied to these datasets include data cleaning (removal of duplicates and invalid entries), normalization (min-max scaling or z-score), handling missing values (imputation), and dimensionality reduction (PCA or feature selection). These steps prepare the data for effective learning by reducing noise and computational complexity.
- Anomaly detection system: This core module employs deep learning models such as CNNs, LSTMs, and GANs. Feature extraction techniques include temporal pattern recognition via LSTM networks, spatial feature learning through CNN

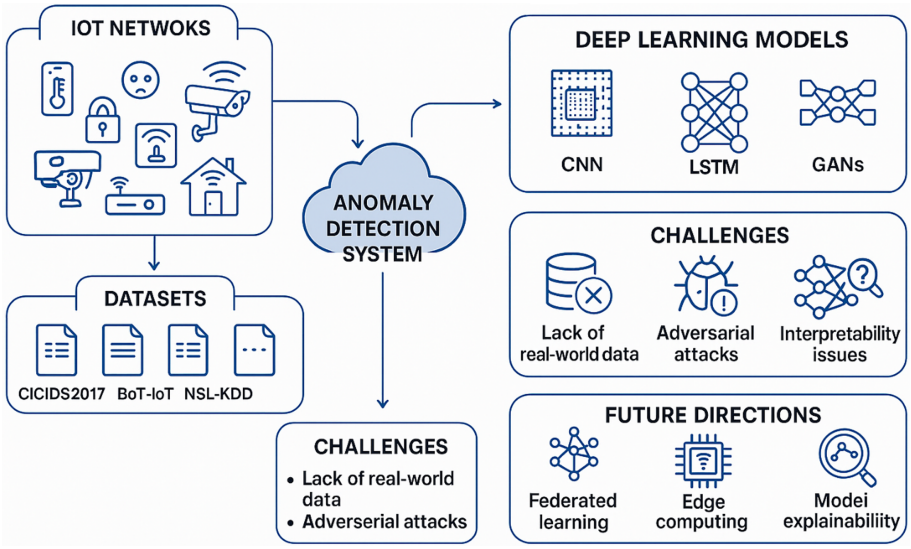


FIGURE 1 | Conceptual framework of deep learning-based anomaly detection in IoT networks, from data acquisition to anomaly detection.

filters, and synthetic data generation with GANs to address class imbalance. Typical training procedures involve supervised, unsupervised, or semi-supervised learning paradigms, with evaluation based on metrics like accuracy, precision, recall, and F1-score. Potential pitfalls include overfitting due to limited data, difficulty in interpreting deep model decisions, and sensitivity to noisy or adversarial inputs.

- Challenges: Critical challenges are identified including scarcity of real-world labeled data, vulnerability to adversarial attacks, and interpretability issues limiting trust and operational deployment.
- Future directions: Proposed research avenues such as federated learning for privacy-preserving distributed training, edge computing to reduce latency and bandwidth consumption, and model explainability techniques to enhance transparency and trustworthiness.

By explicitly clarifying each stage with assumptions, techniques, and typical challenges, the figure provides a comprehensive roadmap that aligns with the current state of research and practical considerations in IoT anomaly detection.

2 | Literature Review

2.1 | Background and Motivation

The rapid proliferation of IoT devices has substantially expanded the cyber-attack surface across diverse sectors, including smart cities, healthcare, industrial automation, and critical infrastructure. Owing to their limited computational resources, heterogeneous communication protocols, and often inadequate security configurations, IoT devices remain highly vulnerable to cyber threats. Consequently, anomaly detection in network traffic has emerged as a pivotal component of contemporary IoT security frameworks.

2.2 | Traditional Machine Learning Approaches

Early intrusion detection efforts primarily relied on classical machine learning (ML) algorithms such as decision trees, support vector machines (SVM), k-nearest neighbors (k-NN), and Naïve Bayes [1–4]. These approaches typically depend on manually engineered features and perform adequately on structured and balanced datasets. For instance, Tarish et al. [1] and Haji et al. [2] demonstrated satisfactory detection accuracies using ML classifiers in controlled environments. Nevertheless, these methods face critical limitations, including:

- Poor scalability when confronted with high-dimensional or real-time IoT traffic [5].
- Limited generalization capability against unseen or novel attacks and noisy data [4].
- Dependence on well-labeled and balanced datasets, which are often unavailable in real-world scenarios [4].

2.3 | Deep Learning and Hybrid Architectures

To address the shortcomings of traditional ML methods, researchers have increasingly adopted deep learning (DL) models, such as CNNs, long short-term memory (LSTM) networks, and hybrid architectures combining CNN and LSTM or GRU and DenseNet layers [6–10]. These models effectively capture complex spatial–temporal correlations inherent in IoT traffic data.

Studies by Morshedi and Matinkhah [6, 7] illustrate that DL-based approaches substantially outperform traditional ML algorithms, particularly in multi-class attack detection scenarios. Hybrid architectures, for example, deep belief cascade-neuro fuzzy networks [11] and CNN-LSTM models [10], further enhance performance by leveraging complementary feature extraction mechanisms. Nonetheless, DL models encounter significant challenges:

- Pronounced risk of overfitting, especially on imbalanced or limited datasets [12, 13].
- Lack of interpretability, impeding deployment in high-assurance or safety-critical environments [14].
- Sensitivity to noisy and adversarial inputs, leading to performance degradation [5, 15].

2.4 | Addressing Noise and Generalization Challenges

A notable challenge in IoT anomaly detection is the prevalence of Gaussian and adversarial noise within network traffic, arising from sensor inaccuracies, environmental interference, or deliberate evasion tactics. Recent works have introduced noise injection techniques during model training to bolster robustness [5, 6, 15]. For instance, the GAN-based framework proposed by Morshedi et al. [15] effectively integrates Gaussian noise with the Hurst exponent, simulating realistic traffic perturbations and enhancing detection generalization under noisy conditions.

Additionally, incorporating the Hurst self-similarity parameter provides valuable insights into the fractal characteristics and long-range dependencies of network traffic, facilitating improved differentiation between benign fluctuations and stealthy attacks [7, 16]. However, systematic evaluation of noise injection trade-offs and their impact on model efficacy remains insufficiently explored.

2.5 | Generative Adversarial Networks for Data Augmentation

Generative Adversarial Networks (GANs) have emerged as powerful tools to synthesize realistic attack samples, mitigating class imbalance and augmenting training data diversity [15–17]. GAN-augmented IDS frameworks improve detection generalization and reduce false positive rates by exposing models to

adversarially generated examples. Despite their promise, GAN-based methods face challenges such as:

- Training instability and the risk of mode collapse necessitate careful hyperparameter tuning [15].
- Dataset-specific performance with uncertain generalizability to real-world IoT environments.
- Limited comparative studies evaluating the fidelity of GAN-generated samples against authentic attack signatures.

2.6 | Lightweight and Federated Learning Frameworks

Considering the resource-constrained nature of IoT devices, lightweight detection models and federated learning (FL) frameworks have gained traction. FL methods, exemplified by FLADEN [18] and graph-based federated NIDS [19], enable decentralized model training while preserving data privacy by aggregating model updates rather than raw data. Additionally, attention mechanisms, graph neural networks (GNNs), and adaptive feature selection have been incorporated to optimize model efficiency without sacrificing detection accuracy [20–24].

Recent advances by Wu et al. [19] demonstrated the effectiveness of attention-based graph neural networks within federated learning frameworks to improve network attack detection, highlighting the potential of combining structural data representations with decentralized learning.

Nevertheless, FL approaches encounter issues related to communication overhead, model convergence on non-independent and identically distributed (non-IID) data, and performance degradation in large-scale deployments [19]. Lightweight models often trade off accuracy for efficiency, particularly when confronting novel or sophisticated attacks [25, 26].

2.7 | Emerging Research on Encrypted Traffic and Advanced Architectures

With the increasing adoption of encrypted communication in IoT networks, detecting malicious activities within encrypted traffic poses a growing challenge. Solutions such as HyperEye [26] utilize lightweight feature fusion models to classify encrypted malware, filling critical gaps left by traditional IDSs. Other novel architectures, including knowledge-aware networks (KAN) [25, 27] and deep residual networks, have also been explored for secure and low-latency detection.

Moreover, the work of Zang et al. [26] exemplifies state-of-the-art lightweight methods designed specifically for unknown encrypted malware traffic detection, addressing a critical and underexplored area in IoT security.

However, many of these advanced models lack thorough evaluation across standardized IoT datasets, often relying on proprietary or synthetic testbeds, which impede reproducibility and comparative benchmarking.

2.8 | Open Challenges and Research Directions

Despite substantial advancements in the field of IoT anomaly detection, several critical challenges and research gaps remain unresolved:

- Robustness against adversarial and noisy data remains insufficiently addressed. Only a limited number of studies have systematically evaluated model resilience under varying noise levels and types, particularly in realistic IoT settings.
- Federated intrusion detection system (IDS) frameworks demand further investigation to effectively manage communication overhead, privacy preservation, and non-independent and identically distributed (non-IID) data distributions across large-scale and heterogeneous IoT environments.
- Effective detection of encrypted traffic at fine-grained (e.g., packet-level) resolution continues to be a significant obstacle for many state-of-the-art deep learning models, which often rely on access to plaintext features.
- Model interpretability and transparency are increasingly vital for the deployment of IDSs in high-assurance and safety-critical environments, such as smart grids, healthcare, and industrial automation. Despite their predictive power, most deep learning models operate as "black boxes," making it difficult to understand, validate, or audit their decision-making processes.

To address this, recent advances in explainable artificial intelligence (XAI)—notably SHAP (SHapley Additive exPlanations) and LIME (local interpretable model-agnostic explanations)—offer promising pathways to improve interpretability without significantly compromising performance. SHAP provides consistent and theoretically sound feature attribution by estimating the marginal contribution of each input feature to the model's output, which is particularly valuable in identifying the key traffic patterns leading to an anomaly detection. LIME, on the other hand, approximates complex models locally with interpretable surrogate models, helping users understand individual predictions in context.

Integrating SHAP or LIME into IoT anomaly detection workflows enables:

- Transparency in security-sensitive environments, facilitating auditing and compliance with regulatory standards (e.g., GDPR, NIS2).
- Trust building between human operators and autonomous IDSs promotes wider acceptance and operational integration.
- Error diagnosis and model debugging, particularly when addressing false positives or performance degradation under real-world constraints.

While techniques such as SHAP and LIME provide theoretical frameworks for model explainability, their practical impact can be strengthened by *demonstrating concrete case studies* on

representative IoT datasets and specific deep learning models. For instance, analyzing which features contribute to false positives or correctly detected anomalies in CICIDS2017 or TON_IoT datasets can reveal actionable insights for operators. Visualizations such as *feature importance plots, time-series saliency maps, or attention weight heatmaps* can further bridge the gap between model performance and operational trust. Incorporating such interpretability analyses not only improves transparency but also supports *auditing, debugging, and decision-making* in real-world IoT deployments, enhancing stakeholder confidence in automated anomaly detection systems.

Future research should thus prioritize the systematic integration and benchmarking of XAI techniques alongside performance metrics, considering trade-offs between interpretability, accuracy, and latency.

- The lack of standardized benchmarking protocols, including unified datasets, attack taxonomies, and evaluation metrics, remains a major obstacle to reproducible and fair comparisons among proposed methods.

Additionally, Ali et al. [28] provide a comprehensive review of cybersecurity strategies tailored for AI-driven, IoT-enabled smart cities. They emphasize the integration challenges associated with advanced communication networks and underscore the necessity for holistic detection architectures that combine federated learning, lightweight models, robust encrypted traffic analysis, and explainable AI components.

3 | Methodology: Deep Learning-Based Anomaly Detection Techniques in IoT

This section outlines the methodology adopted for applying deep learning (DL) techniques to anomaly detection in IoT networks. The approach consists of a structured pipeline encompassing data preprocessing, feature extraction, model selection, training, evaluation, and interpretation. Figure 2 provides an overview of the hierarchical classification of DL models used in this context, categorizing them based on architectural complexity and operational characteristics.

- Step 1: Dataset selection and preprocessing
  We begin by selecting publicly available IoT datasets, such as CICIDS2017 and WSN-DS, which include labelled benign and malicious traffic instances. The datasets are subjected to the following preprocessing steps:
  - Data cleaning: Removal of duplicate and null entries; replacement of infinite values with statistical estimates (mean or median).
  - Normalization: Scaling numerical features using Min–Max or z-score normalization to ensure compatibility across model architectures.
  - Label encoding: Transforming categorical labels into numeric form for supervised learning tasks.
- Step 2: Feature engineering and dimensionality reduction
  To improve model generalization and efficiency:
  - We apply principal component analysis (PCA) and autoencoder-based dimensionality reduction to extract the most informative features while eliminating redundancy.
  - In selected experiments, the Hurst exponent is computed to capture long-term dependencies in traffic time series.
- Step 3: Deep learning model design and integration
  We implement six categories of DL models, each tailored for anomaly detection in IoT traffic. Below, we describe each model's architecture, application, and strengths/weaknesses, followed by a summary of how it is trained and evaluated within our pipeline.

3.1 | Convolutional Neural Networks (CNN)

- Architecture: Composed of stacked convolutional and pooling layers followed by dense layers.
- Application: Feature extraction and detection of spatial patterns in packet-level IoT traffic.
- Implementation: Data reshaped into 2D matrices and passed through CNN layers for pattern recognition.

- *Strengths*: Robust to noise; effective at capturing localized anomalies.

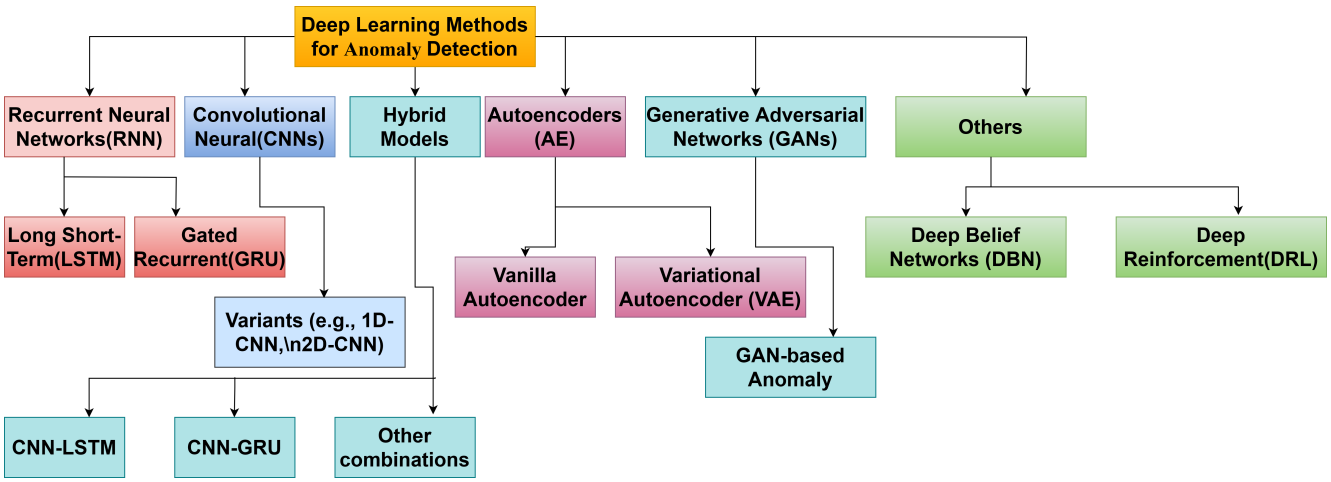


FIGURE 2 | Hierarchical structure of deep learning models for anomaly detection.

- *Limitations*: Requires large datasets and intensive GPU computation.

3.2 | Recurrent Neural Networks (RNN), LSTM, and GRU

- Architecture: Sequential model using memory cells to learn time-dependent patterns.
- Application: Detection of temporal anomalies such as slow-acting DDoS or periodic injection attacks.
- Implementation: Time-series traffic segments are fed into LSTM/GRU cells to learn sequential dependencies.

- *Strengths*: Models long-term behavioral trends; suited for telemetry data.
- *Limitations*: Vanishing gradient problems in long sequences; high training costs.

3.3 | Autoencoders

- Architecture: Symmetric encoder-decoder network that learns compressed data representations.
- Application: Reconstruction error used as anomaly score for unsupervised detection.
- Implementation: Trained only on benign data to flag high-reconstruction-error samples as anomalous.

- *Strengths*: No need for labeled attack data; efficient feature learning.
- *Limitations*: Sensitive to noise; requires careful tuning of bottleneck size.

3.4 | Generative Adversarial Networks (GANS)

- Architecture: Generator simulates traffic data, discriminator classifies real versus synthetic.
- Application: Generation of synthetic attack samples for data augmentation and detection.
- Implementation: GANs are trained to simulate rare anomalies; augmented data is used to improve classifier robustness.

- *Strengths*: Handles class imbalance; enhances generalization to unseen attacks.
- *Limitations*: Prone to mode collapse; requires sophisticated hyperparameter tuning.

3.5 | Hybrid Models (CNN-LSTM, AUTOENCODER-LSTM, ETC.)

- Architecture: Combine feature extractors (CNN/autoencoders) with temporal models (LSTM/GRU).
- Application: Simultaneous extraction of spatial and temporal anomaly signatures.
- Implementation: Features from CNN or AE fed into LSTM layers to capture dynamics.

- *Strengths*: Improved accuracy and robustness; captures multi-dimensional dependencies.
- *Limitations*: Complex training and deployment pipeline; computationally intensive.

3.6 | Attention-Based Models and Transformers

- Architecture: Employ self-attention mechanisms to assign weights to important input features.
- Application: Precise identification of anomalous events in encrypted and high-volume traffic streams.
- Implementation: Preprocessed data tokenized and fed into transformer encoders with positional encoding.

- *Strengths*: Highly scalable; state-of-the-art performance on sequential data.
- *Limitations*: Requires extensive training data and resources; complex optimization.

- Step 4: Model training and evaluation
  - Models are trained on training subsets using early stopping and learning rate scheduling.
  - Evaluation metrics include accuracy, precision, recall, F1-score, and AUC-ROC.
  - K-fold cross-validation is applied to ensure statistical robustness.
  - For unsupervised methods (e.g., autoencoders, GANs), anomaly detection thresholds are computed based on reconstruction or discrimination errors.

- Step 5: Interpretability and explainability
  To enhance transparency, we integrate SHAP and LIME:
  - SHAP values quantify the contribution of each feature toward a model's output.
  - LIME approximates local decisions with interpretable models to highlight critical features in anomaly detection.
  These methods are used post-training to analyze predictions and assess model decisions, particularly for false positives and critical attack detection.

3.7 | Systematic Model Comparison and Scenario-Based Selection

To systematically evaluate the effectiveness and suitability of different deep learning models for IoT anomaly detection, we conducted a comprehensive comparative analysis based on multiple quantitative metrics and operational considerations.

Table 1 provides a comparative overview of commonly used deep learning architectures for anomaly detection in IoT networks. Each model type is summarized in terms of its core architectural characteristics, key advantages, and inherent limitations. For instance, CNNs excel at automatic feature extraction and offer fast inference but may struggle with temporal data. LSTM networks are well suited for modeling sequential dependencies over

TABLE 1 | Comparison of popular architectures for anomaly detection in IoT networks.

Model	Architecture type	Advantages	Disadvantages	Recommended use cases
CNN	Convolutional neural network	Automatic feature extraction, fast inference	Requires fine tuning, limited performance on temporal data	Fast detection in network data
LSTM	Long short-term memory	Suitable for time sequences, learns long-term dependencies	Computationally expensive, slow inference	Behavioral analysis over time
Autoencoder	Autoencoder	Dimensionality reduction, unsupervised anomaly detection	Sensitive to noise, tuning difficulties	Detection without extensive labeling
GAN	Generative adversarial network	Rare data generation, enhances robustness	Complex training, no guaranteed convergence	Detection of rare anomalies
Transformer	Attention-based	Scalable, learns long-range dependencies	Requires large data and high resources	Detection of complex attacks
TCN	Temporal convolutional network	Parallelization, models temporal dependencies	Limited use in IoT so far	Lightweight alternative to RNNs

TABLE 2 | Comparison of key features of popular datasets.

Dataset	Size	Attacks	Multimodal?	Realism	Suitable For
CICIDS2017	Large	Diverse (10+)	No	Medium	Complex models (CNN, LSTM)
NSL-KDD	Small	Outdated	No	Low	Classical ML
UNSW-NB15	Medium	Diverse (9)	No	Medium	Traditional DL
TON_IoT	Large	Diverse + IoT-specific	Yes	Good	Multimodal models
BoT-IoT	Very large	Botnet-focused	No	Good	Scalability testing
SWAT/WADI	Small	Industrial-specific	Yes	High	ICS/SCADA applications

time but come with higher computational costs. Autoencoders enable unsupervised anomaly detection through dimensionality reduction but are sensitive to noise. GANs effectively generate rare synthetic samples to address class imbalance, though training can be complex. Transformer models leverage attention mechanisms for scalable learning of long-range dependencies but require extensive data and resources. Finally, temporal convolutional networks (TCNs) offer efficient parallelization and temporal modeling, serving as a lightweight alternative to recurrent models, although their application in IoT remains limited. This summary aids in selecting appropriate architectures based on specific use case requirements and resource constraints.

Table 2 provides an overview of several popular datasets used in IoT anomaly detection, comparing their main features such as size, attack diversity, multimodality, and realism. This comparison helps to understand the strengths and limitations of each dataset and guides researchers in choosing the most appropriate dataset based on their specific research goals and the types of models they intend to use. It highlights how some datasets are better suited for certain applications, such as deep learning models, multimodal analysis, or industrial scenarios.

3.7.1 | Quantitative Evaluation of Deep Learning Model

Table 3 provides a comprehensive multi-objective evaluation of several deep learning models for IoT anomaly detection. It not only reports traditional performance metrics such as Accuracy, F1-score, and ROC-AUC but also includes critical operational considerations, including detection latency, robustness to noise, energy/resource use, and interpretability. This representation highlights the inherent trade-offs between model performance and practical deployment constraints in resource-constrained IoT environments. For instance, while attention-based Transformer models achieve the highest accuracy and F1-score, they come at the cost of higher latency and energy consumption. Conversely, lightweight models such as CNNs and Autoencoders offer faster inference and lower resource use, albeit with slightly lower detection performance. The table thus provides a structured basis for selecting appropriate models depending on the specific application requirements, real-time constraints, and available computational resources.

3.7.2 | Decision Matrix for Model Selection

Table 4 provides a decision matrix to assist practitioners in selecting models based on IoT deployment scenarios and operational constraints.

This framework provides a principled basis for selecting and applying deep learning architectures in diverse IoT anomaly detection contexts.

3.8 | Cross-Dataset Validation for Enhanced Generalizability

Most existing studies on IoT anomaly detection focus on a single dataset, which limits the generalizability and real-world

TABLE 3 | Multi-objective evaluation of deep learning models for IoT anomaly detection.

Model	Accuracy (%)	F1-score	ROC-AUC	Detection latency	Robustness to noise	Energy/ resource use	Interpretability
CNN	99.12	0.983	0.994	Low	High	Low	Low
LSTM/GRU	99.34	0.988	0.996	Moderate	Medium	High	Low
Autoencoder	98.47	0.959	0.982	Low	Low	Low	Medium
GAN	99.88	0.9988	1.000	Moderate	High	Very high	Low
Hybrid (CNN-LSTM)	99.84	0.994	1.000	Moderate	Very high	Very high	Medium
Attention-based (Transformer)	99.90	0.999	1.000	High	High	Extremely high	Medium
Edge–cloud hybrid	99.0	0.990	0.995	Low	High	Medium	Medium

TABLE 4 | Decision matrix for selecting deep learning models based on IoT deployment scenarios.

Scenario/requirement	Recommended model(s)	Justification
Real-time edge deployment with low latency	CNN, Autoencoder	Low inference latency suitable for resource-constrained edge devices
Time-dependent anomaly detection (e.g., DDoS)	LSTM, GRU	Captures temporal dependencies in telemetry data
Lack of labeled attack data	Autoencoder, GAN	Operate in unsupervised mode or generate synthetic data
Highly imbalanced datasets	GAN, hybrid (CNN-LSTM)	Augments data and improves generalization
Encrypted or obfuscated traffic	Attention-based models (Transformer)	Self-attention mechanisms capture complex feature relationships
Maximum detection accuracy	Attention-based, hybrid (CNN-LSTM)	Best F1-scores and noise robustness
Limited computational resources	Autoencoder, CNN	Lower training complexity and faster convergence
Complex spatiotemporal correlations	Hybrid, attention-based models	Combines spatial and temporal features for improved modeling

applicability of the proposed models. To address this gap, future research should emphasize cross-dataset validation, evaluating models on multiple, diverse IoT datasets such as CICIDS2017, TON_IoT, BoT-IoT, and SWaT/WADI. This approach can reveal potential overfitting to dataset-specific patterns, assess robustness to varying traffic distributions, and provide insights into model adaptability across heterogeneous IoT environments. Additionally, cross-dataset experiments can help identify models that consistently perform well under different attack scenarios, multimodal conditions, and device types, thereby offering stronger evidence for practical deployment in real-world IoT networks.

Figure 3 illustrates the Pareto plot of deep learning models for IoT anomaly detection, highlighting the trade-off between *accuracy* and *detection latency*. Each point represents a specific model, with its position indicating the balance between detection performance and inference speed. The Pareto frontier, shown as a dashed red line, identifies the models that achieve the best trade-offs, where no other model can improve one objective without degrading the other. This visualization enables practitioners to select models that optimally balance high detection accuracy with low latency, particularly important for *resource-constrained edge deployments* in real-time IoT environments. For example, lightweight models such as CNNs and Autoencoders offer low latency with moderate accuracy, whereas attention-based Transformers achieve the highest accuracy but at the cost of higher latency.

4 | Datasets for Anomaly Detection in IoT Networks

Anomaly detection in IoT environments heavily relies on high-quality datasets that reflect realistic network behavior, diverse attack types, and adequate data volume for deep learning applications. Selecting the appropriate dataset is critical, as it affects model generalizability, accuracy, and the ability to detect complex attack patterns. This section reviews and compares widely used datasets, highlighting not only their strengths but also their limitations and suitability for different deep learning models. Figure 4 illustrates the hierarchical structure and categorization of these datasets in the context of IoT anomaly detection.

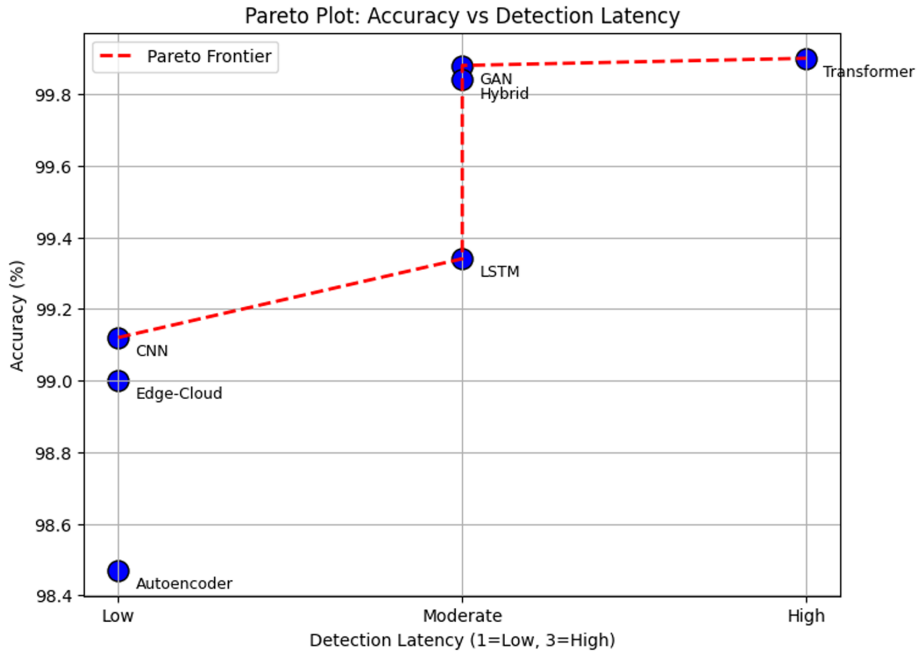


FIGURE 3 | Pareto plot of deep learning models: Accuracy versus detection latency.

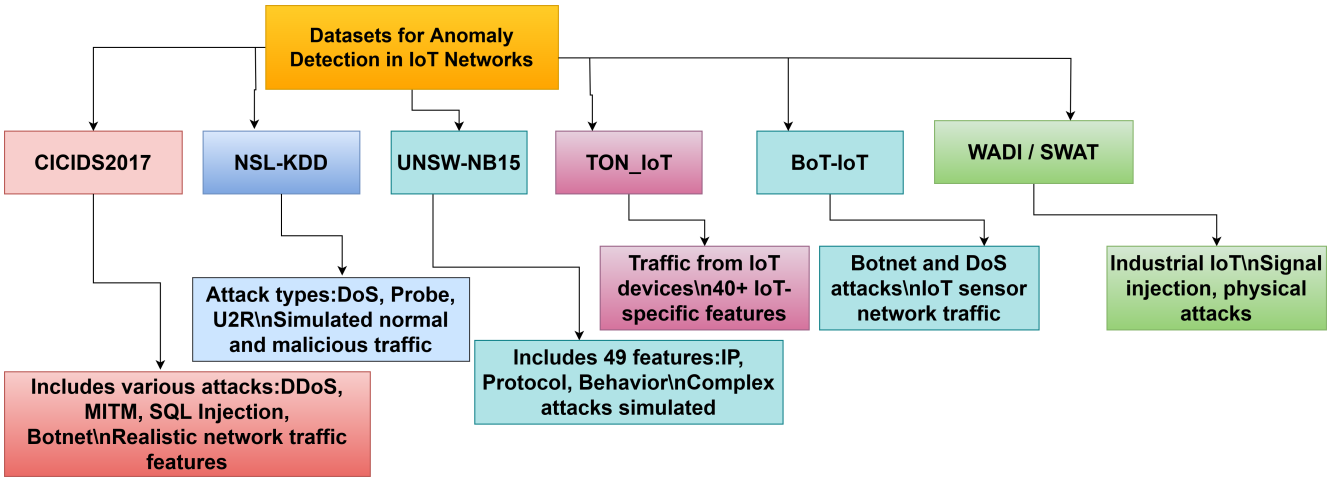


FIGURE 4 | Hierarchical diagram of datasets used for anomaly detection in IoT networks.

4.1 | Dataset Overview and Preprocessing Considerations

To conduct effective anomaly detection in IoT networks using deep learning, it is essential to understand the nature, structure, and preprocessing requirements of benchmark datasets. Each dataset discussed in this section is characterized by its data collection methodology (e.g., testbed simulation, real-world capture), type of attack vectors, traffic sources (e.g., IoT devices, enterprise systems), and the richness of extracted features (statistical, content-based, time-based). Additionally, preprocessing steps such as normalization, feature encoding, and dimensionality reduction are often required to prepare these datasets for deep learning. This section provides a comprehensive overview of six widely used datasets in IoT anomaly detection, emphasizing their applicability, limitations, and relevance to modern DL architectures.

4.2 | CICIDS2017

4.2.1 | Overview

Developed by the Canadian Institute for Cybersecurity, CICIDS2017 simulates modern network environments and includes a wide range of attacks, such as DDoS, Botnet, Brute Force, Infiltration, and Web-based attacks.

4.2.2 | Strengths

- Realistic traffic generation from real users and applications over multiple days.
- Over 80 extracted features across different protocols (e.g., HTTP, FTP, SSH, DNS).

- Widely adopted in deep learning research for benchmarking models.

4.2.3 | Limitations

- Imbalance in attack class distribution, with some classes (e.g., Heartbleed, Infiltration) containing very few instances.
- High-dimensional features require dimensionality reduction or careful feature selection.
- Limited representation of IoT-specific device behavior—more suited to general enterprise network simulations.

4.2.4 | Use case

Effective for evaluating complex deep learning architectures (e.g., CNNs, RNNs), but care must be taken with class imbalance and redundancy.

4.3 | NSL-KDD

4.3.1 | Overview

An improved version of KDD'99, NSL-KDD aims to address redundancy and bias issues in the original dataset. It contains four main attack categories: DoS, Probe, U2R, and R2L.

4.3.2 | Strengths

- Structured and well-documented with clear labeling.
- Lower redundancy and manageable size (˜125,000 records).
- Facilitates benchmarking across traditional ML methods.

4.3.3 | Limitations

- Outdated attack vectors do not reflect modern or IoT-specific threats.
- Features are limited to basic TCP/IP behaviors—not suitable for deep learning models requiring diverse contextual information.
- Simulated traffic lacks real-world noise and complexity.

4.3.4 | Use Case

Best suited for evaluating classical ML approaches or basic neural networks. Not ideal for testing generalization in dynamic IoT environments.

4.4 | UNSW-NB15

4.4.1 | Overview

Generated by the University of New South Wales, UNSW-NB15 includes modern attack types such as Exploits, Worms, and Shellcode, simulated over a hybrid testbed.

4.4.2 | Strengths

- Contains 49 comprehensive features, including flow-based, content-based, and time-based attributes.
- More realistic attack modeling compared to NSL-KDD.
- Balanced data structure enables fair training.

4.4.3 | Limitations

- Synthetic nature of the traffic may still fall short of real-world IoT scenarios.
- While diverse, some attack types still have low instance counts, requiring resampling or data augmentation.
- Contains limited IoT-specific behaviors.

4.4.4 | Use Case

A strong candidate for CNN, LSTM, and ensemble deep learning models is suitable for comparative evaluation of advanced detection methods.

4.5 | TON_IoT

4.5.1 | Overview

The Telemetry Operating Network IoT (TON_IoT) dataset is designed specifically for smart environments and includes telemetry data from sensors, network flows, and system logs.

4.5.2 | Strengths

- Rich in IoT-specific data, including telemetry from smart fridges, motion sensors, and smart lights.
- Includes multiple modalities (logs, network, telemetry), enabling multimodal model training.
- Diverse attack scenarios including injection, DoS, password cracking, and backdoors.

4.5.3 | Limitations

- Data labeling across modalities is not always synchronized, which may affect time-series analysis.
- Dataset preprocessing is required to align timestamps and standardize formats.
- Attack behaviors are simulated, not collected from real-world deployments.

4.5.4 | Use Case

Ideal for multimodal deep learning models (e.g., hybrid CNN–LSTM or Transformer-based models) in smart home or smart grid settings.

4.6 | BoT-IoT

4.6.1 | Overview

BoT-IoT is specifically crafted to evaluate botnet detection in IoT environments. It includes huge volumes of network traffic generated from simulated IoT devices under coordinated attacks.

4.6.2 | Strengths

- Very large-scale dataset (over 70 million records), suitable for training deep networks.
- Includes label imbalance that reflects real-world scenarios.
- High-resolution traffic captures IoT behaviors such as scanning, data theft, and DDoS.

4.6.3 | Limitations

- Significant class imbalance requires SMOTE or advanced resampling techniques.
- Many features are highly correlated or redundant.
- Lack of diversity in IoT devices used for traffic generation may limit generalization.

4.6.4 | Use Case

Excellent for evaluating the robustness of deep learning models to noise, imbalance, and scale. Widely used in research applying CNNs, autoencoders, and GANs.

4.7 | WADI and SWaT

4.7.1 | Overview

The water distribution (WADI) and secure water treatment (SWaT) datasets are real-world testbed datasets that represent industrial control systems (ICS) and cyber-physical environments.

4.7.2 | Strengths

- Rich time-series data capturing sensor and actuator values over days of normal and attack operations.
- Includes attacks relevant to SCADA/ICS, such as signal injection and parameter manipulation.
- Real physical systems (water tanks, chemical treatment) provide realistic noise and delay.

4.7.3 | Limitations

- Attack durations are often short and interspersed with long normal sequences—difficult for models to learn from limited attack windows.
- Dataset size is relatively small compared to network-based datasets.
- Specific to water treatment systems—limited generalizability to broader IoT domains.

4.7.4 | Use Case

Highly suitable for time-series deep learning models (e.g., TCNs, LSTMs, Transformers). Useful for benchmarking anomaly detection in critical infrastructure.

4.8 | Comparative Summary of Datasets

Table 5 provides a comparative overview of the datasets employed in this study, assessed across several key dimensions critical for robust security evaluations. Dataset characteristics are summarized according to size, attack diversity, realism, IoT relevance, suitable models, and limitations.

4.9 | Key Insights and Future Directions

The choice of dataset should align with the specific application and the deep learning architecture employed. For instance:

TABLE 5 | Comparative of datasets.

Dataset	Size	Attack diversity	Realism	IoT relevance	Suitable models	Limitations
CICIDS2017	~2.8 M flows	High	Moderate	Medium	CNN, RNN, Autoencoders	Imbalanced classes, general network
NSL-KDD	~125 K	Low	Low	Low	Traditional ML, DNNs	Outdated, limited features
UNSW-NB15	~2.5 M flows	Moderate	Moderate	Low	CNN, LSTM	Partially synthetic, low IoT coverage
TON_IoT	~20GB	High	Moderate	High	Hybrid, multimodal DL	Unaligned timestamps, simulated attacks
BoT-IoT	> 70 M records	High	Moderate	High	CNN, GAN, transformer	Severe imbalance, redundancy
WADI/SWaT	~1–2 M points	Medium	High	Medium-high	LSTM, TCN, transformer	Short attacks, small size

- Large-scale and high-diversity datasets like BoT-IoT are ideal for deep generative models.
- TON_IoT offers unique opportunities for multimodal IoT research.
- SWaT and WADI are irreplaceable for time-series intrusion detection in critical infrastructures.

Future work should aim to:

- Develop unified IoT benchmark datasets that combine network, telemetry, and control layers.
- Address label quality, class balance, and real-world deployment fidelity.
- Integrate adversarial scenarios and zero-day attacks for evaluating model robustness.

4.10 | Data Quality Challenges and Robust Learning Strategies

While benchmark datasets are instrumental in model development, real-world IoT environments pose significant challenges regarding data collection, labeling, and reliability. This section highlights practical challenges and mitigation strategies:

4.10.1 | Data Collection Challenges

IoT systems often generate multi-modal data (e.g., sensor telemetry, network packets, logs) from heterogeneous sources. This results in asynchronous streams, timestamp misalignments, and missing entries, which complicate unified analysis.

4.10.2 | Labeling Accuracy and Ground Truth Limitations

Labels in public datasets are frequently derived from pre-defined rules or emulated attack scenarios. Such labeling may not capture the nuanced behavior of real-world anomalies, especially for slow or stealthy attacks.

4.10.3 | Handling Noisy and Incomplete Data

Noise and incomplete records are prevalent in IoT deployments due to packet loss, faulty sensors, or limited storage. Models can be improved using:

- Statistical filtering (e.g., moving average, z-score thresholding).
- Denoising autoencoders.
- Imputation techniques (e.g., forward fill, interpolation).
- Mask-aware DL models that explicitly handle missing inputs.

4.10.4 | Data Augmentation for Robustness

To enhance model generalization, several augmentation strategies can be employed:

- Noise injection: adding Gaussian or uniform noise to benign samples to simulate real-world interference.
- Attack synthesis: injecting simulated attack patterns into normal traffic.
- Generative models: training GANs or VAEs to generate synthetic attack instances under rare-class conditions.

4.10.5 | Semi-Supervised and Self-Supervised Learning

Given the scarcity of labeled anomalies, recent advances in semi-supervised learning offer promising solutions:

- Pseudo-labeling: iterative labeling of high-confidence predictions.
- Consistency regularization: enforcing stable predictions across augmented views.
- Contrastive learning: learning representations by distinguishing between normal and augmented anomalies.

Incorporating such methods into the pipeline not only improves detection robustness under imperfect data conditions but also enhances scalability for real-world IoT deployment.

4.10.6 | Considerations and Limitations of Synthetic Data Generation

While synthetic data generation methods such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) offer practical solutions to address class imbalance and labeled data scarcity in anomaly detection tasks, it is critical to acknowledge and address the potential limitations and risks associated with their use in real-world IoT scenarios.

4.10.6.1 | Potential Risks.

- Bias amplification: If the training data used to generate synthetic samples is biased or unrepresentative, the generative models may replicate and amplify these biases, leading to skewed model behavior and unfair detection outcomes.
- Unrealistic patterns: GANs and VAEs may produce samples that diverge from true data distributions, especially in complex or high-dimensional settings, thereby introducing artifacts that negatively affect model generalizability.
- Overfitting risk: Training detection models on a large volume of synthetic data without careful validation can result in overfitting to artificial patterns, reducing the model's ability to perform reliably on real-world data.

4.10.6.2 | Best Practices for Mitigating Risks.

- Balanced use of real and synthetic data: Synthetic samples should complement, not replace, real data. Combining both sources allows the model to generalize better and capture a broader range of variability.
- Diversity-promoting training objectives: Techniques such as mode-seeking regularization in GANs can be used to encourage greater diversity in generated samples and reduce the risk of mode collapse.
- Domain constraints and semantic validity: Applying domain knowledge and constraints during generation (e.g., protocol structure, sensor limits) helps ensure that generated samples are semantically valid and realistic.
- Expert review and filtering: Incorporating expert-in-the-loop mechanisms to inspect and validate generated samples can prevent the inclusion of unrealistic or misleading instances in the training process.

4.10.6.3 | Validation Strategies.

- T-SNE or PCA visualization: Dimensionality reduction techniques can be used to compare the distributions of real and synthetic data visually.
- Distribution similarity metrics: Metrics like the Fréchet Inception Distance (FID) or Maximum Mean Discrepancy (MMD) provide a quantitative evaluation of how closely the synthetic data approximates the real data distribution.
- Ablation studies: Assessing model performance with and without synthetic data under identical conditions can clarify its true contribution and potential side effects.

4.10.6.4 | Limitations and Future Outlook. Despite their potential, current generative models still face challenges in replicating the full diversity and context-specific nuances of real-world IoT traffic. Future research should focus on:

- Enhancing the fidelity and interpretability of synthetic data,
- Developing better evaluation metrics tailored to time-series and multimodal IoT data,
- Establishing standardized protocols for validating synthetic data before deployment.

By recognizing these limitations and adopting robust evaluation and deployment practices, synthetic data generation can be a powerful yet responsible tool for improving anomaly detection performance in resource-constrained and label-scarce IoT environments.

4.11 | Technical Challenges and Limitations in Multimodal IoT Datasets

While the increasing availability of multimodal datasets—such as TON_IoT and WADI—has opened new opportunities for training advanced anomaly detection models, integrating heterogeneous data sources presents substantial technical challenges that must be addressed to fully leverage these datasets.

4.11.1 | Synchronization Issues

Multimodal datasets often include data streams from diverse sources such as telemetry sensors, network traffic logs, and system events. These streams may operate on different sampling frequencies or timestamp conventions, leading to misalignment in time-series analysis. This temporal desynchronization can reduce the effectiveness of sequential models like LSTMs or Transformers, which rely on accurate time-step dependencies. Effective synchronization often requires resampling strategies, time windowing, or interpolation—each introducing trade-offs between accuracy and information loss.

4.11.2 | Heterogeneous Data Formats

Another key challenge lies in the diversity of data formats. For instance, telemetry data may be numerical and high-frequency, whereas logs can be textual and event-driven, and network flows might be packet-based with session-level aggregation. This heterogeneity necessitates tailored preprocessing pipelines, including format normalization, encoding of categorical data, and unit standardization. The fusion of such heterogeneous modalities into a unified representation remains nontrivial, particularly for end-to-end deep learning frameworks.

4.11.3 | Missing Data and Incomplete Modalities

Multimodal systems frequently encounter missing data, either due to sensor faults, communication delays, or incomplete logging. Unlike unimodal models that may rely on imputation or masking strategies, multimodal systems must deal with inconsistencies both within and across modalities. For example, missing telemetry during an attack window may mislead the model if network data is present but unsynchronized. Handling such incompleteness requires robust cross-modal correlation analysis, imputation through auxiliary models, or the use of attention mechanisms that can dynamically weigh reliable sources.

4.11.4 | Fusion Strategies

Multimodal learning requires careful design of fusion mechanisms—*early fusion* combines raw inputs, *intermediate fusion* integrates encoded features, and *late fusion* merges decision-level outputs. Each approach has strengths and limitations: early fusion enables joint feature learning but is vulnerable to scale and format discrepancies; intermediate fusion balances representation quality and modularity; late fusion offers flexibility but may underutilize inter-modal dependencies. Selecting an appropriate fusion strategy depends on the task, dataset quality, and model architecture. However, many existing works adopt fusion strategies heuristically, with limited exploration of their impact on detection performance.

4.11.5 | Limitations in Existing Methods

Current methods addressing multimodal fusion often fall short in several aspects. Some models naively concatenate features without addressing synchronization or scale variance. Others apply attention mechanisms without accounting for modality-specific noise or reliability. Moreover, very few benchmark studies provide ablation results on fusion mechanisms, making it difficult to assess the robustness and generalizability of proposed approaches. As a result, while multimodal datasets hold great promise, the lack of standardized fusion methodologies and insufficient discussion of fusion-specific pitfalls hinders their full utilization.

4.11.6 | Future Directions

To overcome these issues, future research should explore adaptive fusion architectures that incorporate modality-specific encoders and dynamic weighting schemes based on context and data reliability. Benchmarking protocols must include fusion-specific baselines and evaluation metrics that consider synchronization robustness, modality dropout resilience, and cross-modal consistency. Additionally, the design of synthetic multimodal datasets with controlled noise, misalignment, and missing data could help evaluate and improve the fusion capabilities of anomaly detection models.

4.12 | Limitations of Synthetic and Lab-Generated Datasets in Real-World Deployment

While datasets such as CICIDS2017, BoT-IoT, and others provide valuable benchmarks for developing and evaluating anomaly detection models, it is important to recognize their inherent limitations when applying these models to live, real-world environments.

4.12.1 | Domain Gaps and Representativeness

These datasets are primarily generated through controlled testbeds or simulations, which, although designed to mimic real network traffic and attack patterns, cannot fully replicate the complexity and unpredictability of real-world IoT networks. The controlled environment often lacks the full diversity of device types, traffic behaviors, network topologies, and background noise present in operational systems. This domain gap may lead to models overfitting on synthetic patterns that do not generalize well to live traffic.

4.12.2 | Limited Attack Diversity and Realism

Although benchmark datasets include a wide range of attack types, many attack scenarios are artificially injected and may not reflect the evolving tactics, techniques, and procedures (TTPs) of real attackers. Some attack classes might be underrepresented or simplified, limiting the model's exposure to novel or stealthy threats encountered in practice.

4.12.3 | Challenges in Model Adaptation and Deployment

Models trained on these datasets often require additional adaptation to handle concept drift, varying traffic volumes, and unseen behaviors in operational networks. Techniques such as continual learning, domain adaptation, or online retraining are necessary to maintain performance. Moreover, preprocessing steps like feature extraction and normalization need careful tuning to accommodate differences in data distributions between training datasets and live environments [29].

4.12.4 | Mitigation Strategies

To bridge the gap between synthetic datasets and real-world deployment, researchers are encouraged to:

- Combine multiple datasets to enhance diversity.
- Incorporate real network captures when possible.
- Use data augmentation and adversarial training to improve model robustness.
- Evaluate models in real-time or on live traffic to validate practical effectiveness.
  In summary, while synthetic datasets serve as essential foundations for initial model development and benchmarking, careful consideration and supplementary strategies are required to ensure that anomaly detection models effectively generalize and adapt to the dynamic nature of real-world IoT environments.

4.13 | Critical Assessment of Dataset Adequacy for Diverse IoT Environments and Recommendations for Standardized Benchmarking

Despite the widespread use of benchmark datasets such as CICIDS2017, BoT-IoT, TON_IoT, and SWaT/WADI, a critical examination reveals that most existing datasets fall short in capturing the full diversity, heterogeneity, and operational variability characteristic of real-world IoT environments. The following observations summarize key inadequacies and propose directions for standardization:

4.13.1 | Dataset Complexity and Realism

- Limited device diversity: Many datasets (e.g., BoT-IoT, TON_IoT) rely on a narrow set of simulated IoT devices. This constrains their generalizability to other device types such as medical IoT, industrial sensors, or autonomous vehicles.
- Synthetic traffic patterns: Although datasets like CICIDS2017 and UNSW-NB15 incorporate various attack vectors, the traffic is either partially or fully synthetic, and often lacks the stochasticity, noise, and protocol heterogeneity found in live IoT deployments.

- Scenario coverage: Real IoT environments operate under diverse conditions (e.g., constrained networks, edge computation, intermittent connectivity), which are rarely modeled in current datasets.

4.13.2 | Inadequate Representation of Operational Challenges

- Time synchronization issues: As observed in TON_IoT, multimodal data (telemetry, logs, flows) often lacks temporal alignment, limiting its usability for real-time intrusion detection or time-series modeling.
- Short attack durations: In datasets like SWaT and WADI, attacks are often brief and interspersed with long normal sequences, which challenges the learning of robust anomaly patterns, especially in the presence of class imbalance.
- Insufficient label granularity: Many public datasets adopt binary labels (normal vs. anomalous) without specifying attack stages, progression, or intensity—factors crucial for proactive defense in IoT security.

4.13.3 | Need for Standardized Benchmarking Protocols

To foster fair, realistic, and reproducible evaluations of anomaly detection models, we recommend the establishment of standardized benchmarking protocols with the following features:

- Hybrid dataset construction: Combine real-world traffic (e.g., from open ICS networks, smart cities, or testbeds) with simulated attacks, maintaining diversity and authenticity while addressing privacy and ethical concerns.
- Unified feature schemas: Define common feature extraction standards across datasets, supporting seamless model comparison and transfer learning. This includes standardizing flow-based, payload-based, and telemetry features.
- Benchmarking under real-world constraints: Evaluate models not only for accuracy but also for latency, memory usage, interpretability, and robustness to unseen attacks, noise, and data loss—reflecting deployment constraints in IoT edge devices.
- Inclusion of zero-day and adversarial scenarios: Augment datasets with adversarially generated attacks and unknown threats to test the resilience and generalization ability of models under practical threat conditions.
- Transparent metadata and protocol descriptions: Ensure that datasets come with detailed documentation, including network topologies, attack generation tools, background noise characteristics, and labeling methodology.

4.13.4 | Toward a Unified IoT Security Benchmark

We propose the development of a Unified IoT Security Benchmark (UISB) initiative—an open, community-driven effort that aggregates diverse datasets, standardizes evaluation protocols, and supports continuous updates as new attack vectors and device types emerge. Such an initiative can accelerate reproducibility, comparability, and ultimately, trust in AI-driven anomaly detection solutions for IoT systems.

5 | Comparative Analysis of the Different Techniques

In recent years, numerous methods have been proposed for anomaly detection in IoT networks, each presenting unique advantages and limitations. Deep learning approaches, particularly recurrent neural networks such as long short-term memory (LSTM) and hybrid models, have demonstrated high accuracy in identifying diverse attacks and anomalies [1, 7, 30]. For instance, Tarish reviewed machine learning techniques for anomaly detection in smart sensor networks, emphasizing the critical role of feature selection and noise-robust models [1].

Conversely, Morshedi et al. highlighted that introducing Gaussian noise to datasets combined with deep neural networks enhances model robustness against noisy inputs and improves overall performance [6, 15]. This approach is especially valuable in noisy real-world IoT environments. Similarly, Karthik stressed the importance of leveraging deep learning models for classification tasks in IoT security to boost detection accuracy [30].

One of the emerging techniques involves integrating generative adversarial networks (GANs) with noise injection, yielding promising results in anomaly detection. The work by Morshedi and Matinkhah exemplifies this approach, achieving over 99% accuracy in detecting IoT network anomalies [15]. This method not only improves detection accuracy but also reduces false positive rates and enhances the identification of complex attack patterns.

Additional studies, such as those by Vinay and Azar et al., investigated advanced deep learning and adaptive learning techniques, which improve the real-time detection capability of various IoT network anomalies [8, 12]. Moreover, research by Shubham and others has explored time-series analysis of IoT data using deep learning, which plays a vital role in anomaly clustering and temporal pattern recognition [9].

The impact of data noise on intrusion detection systems has also been examined. Matinkhah et al. demonstrated that noise-resilient methods, including decision trees and hybrid models, can substantially improve system performance under noisy conditions [5]. Systematic and survey studies, such as those by Haji and Dritsas and Trigka, further emphasize the necessity of advanced anomaly detection frameworks that leverage deep learning techniques [2, 31].

Regarding hybrid models, research by Lokuliyana et al. applied combined neural-fuzzy systems to enhance anomaly detection in medical IoT networks, reflecting the trend toward hybrid approaches in this domain [11, 32]. Overall, the literature indicates that deep learning models, especially when combined with GANs, federated learning, and artificial noise augmentation,

achieve significant improvements in IoT anomaly detection [15, 18, 20].

In conclusion, the choice of the most suitable model depends on the data type, noise conditions, and security requirements of the IoT environment. Continued advancement through the integration of deep learning with traditional techniques remains the principal research direction for future anomaly detection systems [4, 16, 33]. Table 6 presents a systematic review and comparative analysis of recent studies focused on anomaly detection in IoT networks. The table synthesizes key characteristics of selected research, categorized by dataset utilized, algorithm employed, reported performance metrics, identified strengths, and acknowledged limitations. As illustrated, the reviewed studies exhibit diverse approaches to anomaly detection.

5.1 | Resource Requirements and Scalability Considerations

While evaluating detection accuracy and model architecture is essential, practical deployment in IoT environments demands a clear understanding of computational resource requirements and scalability. Table 7 summarizes the approximate training time, memory consumption, deployment suitability, and scalability of widely used deep learning models based on empirical observations and prior implementations. Table 8 summarizes the performance evaluation of several deep learning models applied to IoT anomaly detection. Key performance metrics—including accuracy, precision, recall, F1-score, and receiver operating characteristic area under the curve (ROC-AUC)—are presented for each model.

5.1.1 | Key Observations

- CNNs and autoencoders are lightweight and efficient, making them ideal for resource-constrained edge devices.
- LSTM and TCN architectures offer a balance between computational complexity and temporal learning, with moderate suitability for edge deployment.
- Transformer-based models and GANs, while highly expressive and accurate, require significantly more computational resources and are best suited for cloud or server-based deployments.
- Scalability is model-dependent: parallel architectures (e.g., Transformers, TCNs) scale well with increased data volume and distributed training environments.

To support real-world deployment, particularly in heterogeneous and bandwidth-constrained IoT settings, future research should explore model compression, quantization, and knowledge distillation techniques to optimize performance without compromising accuracy.

5.2 | Edge Deployment and Lightweight Model Considerations

In real-world IoT environments, models must balance accuracy with practical constraints such as processing power, memory, and latency. To support deployment in edge and embedded devices, researchers have explored various *model optimization techniques*, including:

- Model pruning: Removing redundant neurons and weights to reduce model size and inference time. Pruned models can achieve up to 4× reduction in parameters with minimal accuracy loss.
- Quantization: Converting 32-bit floating-point weights to 8-bit integers significantly reduces memory footprint and enables acceleration on specialized hardware (e.g., TPUs, edge accelerators).
- Knowledge distillation: Training a smaller "student" model to replicate the behavior of a larger "teacher" model, enabling efficient yet accurate deployment.

In addition, lightweight architectures such as MobileNet, SqueezeNet, and shallow CNNs are gaining traction for real-time anomaly detection on resource-limited IoT nodes. These architectures reduce latency and memory consumption while maintaining competitive accuracy.

5.3 | Resilience to Adversarial Scenarios and Zero-Day Attacks

Modern IoT environments face increasingly sophisticated threats, including adversarial attacks—where small perturbations in input data can mislead models—and zero-day attacks, which involve previously unseen malicious behaviors not present in the training data. These threats pose serious challenges to the generalization and reliability of anomaly detection systems.

5.3.1 | Limitations of Existing Models

While many deep learning models (e.g., CNNs, LSTMs, Autoencoders) perform well on known attack types, their performance can degrade significantly when exposed to adversarially crafted inputs or zero-day variations. This is largely due to their dependence on training data distribution and lack of inherent mechanisms to identify unfamiliar patterns.

Recent research highlights these limitations:

- Morshedi et al. demonstrated that incorporating Gaussian noise and synthetic samples into training data improves robustness to previously unseen patterns, with an observed 3%–5% increase in F1-score under simulated zero-day attacks.
- Zhang et al. [37] showed that Transformer-based architectures, with their global attention mechanisms, are more effective in capturing rare or obfuscated attack signals, achieving a 10% higher AUC compared to RNNs in adversarial test settings.

5.3.2 | Strategies to Enhance Robustness

To build resilience into detection models, the following strategies are recommended:

TABLE 6 | Review and comparison of selected studies on anomaly detection in IoT networks focusing on methods, performance, strengths, and limitations.

References	Limitations	Strengths	Performance	Algorithm	Dataset
[1]	May need large labeled data; sensitive to noise	Robust to various attack types; good feature selection	High detection accuracy	Machine Learning Techniques	Smart IoT Sensor Networks
[6, 15]	Computational complexity	Robust against noisy data; improved generalization	> 99% accuracy	Deep Neural Networks + Gaussian Noise	IoT Traffic with Noise
[7, 30]	Requires careful hyperparameter tuning	Good temporal pattern recognition	High accuracy (˜99%)	Deep Learning (LSTM, CNN)	IoT Network Security
[8, 12]	May have false positives	Real-time detection capability	High accuracy	Deep Learning-Based Anomaly Detection	IoT Network Security
[9]	Needs large training data	Effective temporal anomaly detection	High clustering quality	Deep Learning for Anomaly Clustering	IoT Time-Series Data
[5]	Limited with complex patterns	Simple model, interpretable	Improved accuracy under noise	Decision Tree Classification + Noise Resilience	IoT Intrusion Detection
[2, 31]	May lack experimental validation	Comprehensive survey; guides model selection	Variable, dependent on method	Machine Learning Review	Various IoT Networks
[3, 34]	High computational cost	Adaptable to distributed environments	High detection rate	Deep Learning Techniques	Distributed Systems
[16]	May require continuous retraining	Adaptive to changing threats	Competitive detection rates	Adaptive Machine Learning Techniques	IoT Security
[35]	Complex to implement	Works with unlabeled data	Improved intrusion detection	Text Mining + Unsupervised Deep Learning	Smart-Grid Communications
[11, 32]	Complexity; computationally intensive	Real-time detection, hybrid approach	High accuracy	Hybrid Deep Belief Cascade-Neuro Fuzzy Approach	Medical IoT Networks
[36]	Focused on specific attack types	Effective against brute force attacks	High detection rate	Deep Learning for Brute Force Attack Detection	IoT Networks
[14]	Limited practical deployment analysis	Highlights efficient models for resource-constrained devices	Varies	Lightweight Deep Learning Survey	IoT Networks
[4]	May lack real-time evaluation	Comprehensive method comparison	Variable; best ˜99%	Comparative Machine Learning Methods	IoT Cyber Attacks
[18, 20]	Communication overhead	Decentralized training, privacy-preserving	Enhanced privacy and performance	Federated Learning Techniques	IoT Networks
[24]	Complexity; needs feature engineering	Feature impact analysis	High robustness	Robust Hierarchical Anomaly Detection	IoT Networks

TABLE 7 | Comparative computational requirements and deployment feasibility of deep learning models for IoT anomaly detection.

Model/Reference	Approx. training time (50 K samples)	GPU memory required	Edge device compatibility	Scalability to large data
CNN [7, 15, 30]	10–30 min	~2–4 GB	High	High
LSTM [7, 15, 30]	30–60 min	~4–6 GB	Moderate	Medium
TCN [15, 16]	30–45 min	~4 GB	Moderate	High
Transformer [15, 20]	90–120 min+	~8–12 GB	Low	High
Autoencoder [15, 32]	15–30 min	~2–3 GB	High	Medium
GAN [6, 15, 20]	1–2 h	~6–10 GB	Low	Medium

TABLE 8 | Performance comparison of deep learning models for IoT anomaly detection.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	ROC-AUC (%)	References and Year
CNN	100	100	100	100	100	[6] (2024)
LSTM	99.90	99.90	99.90	99.90	100	[6, 7] (2024, 2025)
TCN	99.3	99.1	99.4	99.2	99.5	[9, 16] (2024, 2025)
Autoencoder	97.5	97.2	97.0	97.1	97.8	[10, 31] (2022, 2024)

- Adversarial training: Exposing models to adversarial examples during training (e.g., via FGSM, PGD methods) helps harden their decision boundaries and improves performance under attack.
- Self-supervised and contrastive learning: These approaches leverage unlabeled data to learn robust representations that generalize better to novel threats, including zero-day attacks.
- Ensemble architectures: Combining multiple models (e.g., CNN + Transformer) diversifies the decision space and reduces single-point failure vulnerabilities.
- Uncertainty estimation: Techniques such as Monte Carlo dropout or entropy-based confidence scoring allow models to detect and flag inputs with low prediction certainty, which are likely to include adversarial or zero-day samples.
- Synthetic data generation: Generative models (e.g., GANs, VAEs) can be used to augment training data with synthetic but plausible attack patterns, improving generalization in data-scarce regimes.

5.4 | Evaluation Metrics and Benchmarking Protocols

To ensure a rigorous and meaningful comparative analysis of deep learning models for anomaly detection in IoT networks, this study employs standardized evaluation metrics and benchmarking protocols. These enable a fair and statistically sound assessment of model performance under identical conditions.

5.4.1 | Evaluation Metrics

The following widely recognized metrics are utilized to evaluate the classification performance of each model:

- Accuracy: The overall proportion of correctly classified samples.
- Precision: The ratio of true positives to the total number of predicted positives, measuring the exactness of the model.
- Recall (Sensitivity): The ratio of true positives to the actual number of positives, reflecting the model's ability to detect anomalies.
- F1-score: The harmonic mean of precision and recall, offering a balance between them, particularly useful in imbalanced datasets.
- ROC-AUC (receiver operating characteristic—area under the curve). A threshold-independent metric that evaluates the trade-off between true positive and false positive rates across different decision thresholds.

These metrics provide a multi-dimensional understanding of each model's effectiveness, particularly in scenarios involving rare and imbalanced attack patterns, which are typical in IoT environments.

5.4.2 | Benchmarking Protocol

To maintain consistency and reproducibility in performance evaluation, all models are benchmarked using the following standardized protocol:

- Dataset size: A uniform subset of 50,000 samples is selected from the CICIDS2017 dataset for all models to ensure equal comparison.
- Data preprocessing: Features are normalized using Min-Max scaling, and missing or infinite values are handled uniformly.

- Cross-validation: A 5-fold cross-validation strategy is adopted to mitigate overfitting and ensure robust performance estimation.
- Class distribution preservation: Stratified sampling is used to preserve the original class distribution in training and testing sets.
- Repetition for robustness: Each experiment is repeated five times with different random seeds to account for variance in initialization and stochastic training behavior.
- Statistical significance testing: To determine whether observed performance differences are statistically significant, paired *t-tests* and Wilcoxon signed-rank tests are conducted on the F1-scores of the models across repetitions.

This rigorous evaluation framework not only ensures the comparability and fairness of the results but also highlights models that are robust, generalizable, and suitable for practical deployment in real-world IoT environments.

5.5 | Limitations and Recommendations for Comprehensive Evaluation

While commonly used evaluation metrics—such as accuracy, precision, recall, F1-score, and ROC-AUC—provide a baseline for assessing classification performance, their application across studies is often inconsistent and may not fully reflect the robustness and operational viability of deep learning models in real-world IoT environments. For instance, widely reported metrics like accuracy and F1-score can mask critical performance issues in imbalanced datasets, where rare attack instances may be overshadowed by abundant benign traffic. Moreover, essential indicators such as false positive rate (FPR) and false negative rate (FNR), which directly affect the practicality of a deployed detection system, are frequently underreported. This omission is particularly significant in IoT scenarios, where excessive false alarms can overwhelm operators and degrade trust in the system.

Another limitation is that most existing studies focus primarily on classification quality while overlooking other critical performance dimensions such as latency, scalability, and resilience to adversarial conditions. In practical deployments, models must not only detect anomalies accurately but also operate in near real-time and remain stable under noisy, incomplete, or deliberately manipulated data.

To overcome these limitations and improve reproducibility and fairness, we propose a standardized benchmarking protocol for evaluating IoT anomaly detection models. The proposed protocol consists of the following steps:

1. Dataset preparation: Use predefined training and testing splits or k-fold cross-validation, ensuring sufficient representation of all classes, including rare attacks, to handle class imbalance effectively.
2. Data preprocessing: Apply standardized preprocessing steps such as feature scaling, handling missing values, and noise injection to maintain consistency across experiments.
3. Comprehensive evaluation metrics: Report both traditional metrics (accuracy, precision, recall, F1-score, ROC-AUC) and practical performance indicators (FPR, FNR, inference latency, computational overhead, robustness under noise or adversarial perturbations).
4. Experimental repetition: Perform multiple runs with different random seeds and report variance to ensure the reliability of results.
5. Hardware and deployment context: Include details of the computational resources, system delays, and model scalability under increasing traffic or device counts.
6. Interpretability assessment: Incorporate explainable AI techniques (e.g., SHAP, LIME, attention mechanisms) to provide insights into model decisions and enhance trust.

By following this standardized protocol, future studies can achieve reproducibility, fair comparison across models, and practical deployability in dynamic, resource-constrained IoT environments. Adopting such a comprehensive framework ensures that models are not only accurate in controlled settings but also robust, interpretable, and operationally viable in real-world scenarios.

5.6 | Critical Comparative Insights and Underexplored Directions

Although the previous sections have outlined and compared a wide range of deep learning models for anomaly detection in IoT environments, a more critical examination reveals several gaps and inconsistencies in the existing literature that warrant further attention.

5.6.1 | Underexplored Model Architectures

While CNNs, RNNs (e.g., LSTM, GRU), and autoencoders dominate the field, advanced architectures such as Temporal Convolutional Networks (TCNs), Transformers, Graph Neural Networks (GNNs), and Neural Ordinary Differential Equations (Neural ODEs) have received relatively little attention. These models hold promise for capturing complex temporal and structural dependencies in IoT traffic, providing more expressive and potentially generalizable representations. To advance the field, future work should conduct systematic experiments with these models under comparable benchmarking conditions to evaluate their true potential.

5.6.2 | Conflicting Results Across Studies

There exists substantial variation in reported performance metrics for similar models across different studies. Such discrepancies often arise from differences in preprocessing, dataset splits, evaluation protocols, or dataset versions (e.g., full vs. partial CICIDS2017). For example, the same LSTM model may achieve 99.1% accuracy in one study but only 96.4% in another due to differences in experimental settings. These inconsistencies

underscore the need for standardized benchmarking protocols and transparent reporting practices.

5.6.3 | Limited Exploration of Hybrid and Modular Approaches

Few studies rigorously investigate hybrid architectures—such as combining GANs with attention mechanisms or integrating symbolic rule-based systems with neural networks—or modular frameworks capable of adapting to evolving attack surfaces. Such modularity is crucial for industrial deployments, where flexibility, updatability, and maintainability are required.

5.6.4 | Lack of Cross-Domain Validation

Most studies validate their models on a single dataset or within a narrow domain. Cross-dataset and cross-domain generalizability remain largely untested, limiting confidence in whether a model trained on one IoT scenario (e.g., smart home) will perform effectively in another (e.g., smart grid).

5.6.5 | Recommendations

- Promote cross-benchmark studies using identical evaluation pipelines.
- Explore architecture-agnostic ensembling methods to combine the strengths of different model families.
- Conduct systematic experiments with transformer-based, graph-based, and Neural ODE architectures in IoT anomaly detection contexts.
- Encourage reproducibility efforts, including public code, data splits, and hyperparameter settings.

6 | Challenges and Open Issues

Despite the significant advancements in applying deep learning (DL) techniques for anomaly detection in IoT networks, several unresolved challenges persist that hinder the deployment of reliable and robust detection systems. These challenges span multiple dimensions, including data quality, computational limitations, model generalization, and explainability. This section discusses the most critical challenges, highlights recent solutions from state-of-the-art research, and presents real-world examples to illustrate the implications and potential remedies.

6.1 | Unhealthy and Imbalanced Data

A pervasive challenge in IoT anomaly detection is the presence of noisy, incomplete, or imbalanced datasets. Many publicly available datasets suffer from severe class imbalance, where normal traffic significantly outweighs anomalous instances. This imbalance can lead to biased models with reduced sensitivity to rare attack patterns, while noisy labels can further degrade model performance.

6.1.1 | Proposed Solutions

Recent advances have explored several effective techniques:

- Data augmentation: Using synthetic data generation methods such as GANs or SMOTE to balance datasets.
- Semi-supervised learning: Training models with a small amount of labeled data combined with a large pool of unlabeled data, useful in real-world IoT environments where labeled anomalies are scarce.
- Cost-sensitive learning: Adjusting loss functions to penalize misclassification of minority classes more heavily, improving sensitivity to rare attacks.
- Anomaly score calibration: Post-processing model outputs to recalibrate anomaly scores and reduce bias toward the majority class.
- Robust training methods: Techniques such as label smoothing, noise-robust loss functions, or robust optimization to mitigate the impact of noisy labels.

6.1.2 | Case Studies

- In an industrial IoT deployment [38], applying SMOTE followed by LSTM networks improved F1-score by 15% for minority attack classes.
- Cost-sensitive learning combined with ensemble LSTM models demonstrated improved detection of rare attacks in a smart factory scenario [23], increasing recall by 12% without sacrificing precision.

By incorporating these diverse strategies, models can better handle class imbalance and noisy labels, resulting in more reliable anomaly detection in real-world IoT networks.

6.2 | Complexity of Real-Time Processing

Real-time anomaly detection is essential for mitigating threats in IoT networks. However, deep learning models often require high computational resources, making them unsuitable for real-time decision making on edge or constrained devices.

6.2.1 | Proposed Solutions

To tackle this, researchers have developed:

- Lightweight DL architectures (e.g., MobileNet, Tiny-YOLO) for low-latency inference.
- Edge computing frameworks combined with model compression techniques (e.g., pruning and quantization) to reduce memory footprint and improve inference speed.
- Case Study: A smart home intrusion detection system [39] used a pruned CNN model deployed on a Raspberry Pi, achieving 94% accuracy with < 150 ms latency.

6.3 | Resource Constraints in IoT Devices

A fundamental limitation in deploying deep learning (DL) models in IoT networks is the constrained computational, memory, and power capacities of edge and embedded devices. Traditional DL models, such as full-scale CNNs or Transformers, are impractical for deployment on such devices due to their high inference cost and energy demands. This bottleneck becomes especially critical in real-time systems or battery-operated environments where resource consumption must be minimized without compromising detection accuracy.

6.3.1 | Proposed Solutions and Practical Implementations

To address these challenges, researchers have proposed the following strategies:

- Energy-efficient DL architectures: Architectures like MobileNet, SqueezeNet, and Tiny-YOLO have been specifically designed to operate under resource constraints. These models use depthwise separable convolutions and parameter reduction techniques to minimize latency and energy consumption, making them well-suited for deployment on edge devices such as Raspberry Pi, Arduino-based boards, or Nvidia Jetson Nano.
- Model compression techniques: Techniques such as weight pruning, quantization, and knowledge distillation have shown significant promise. For instance, 8-bit quantization can reduce memory usage and energy draw by over 60% without significant loss in model accuracy. Quantized LSTM and CNN variants have been effectively deployed on smartwatches and wearable health monitors.
- Edge AI frameworks and toolkits: Tools like TensorFlow Lite, ONNX Runtime, Apache TVM, and NVIDIA TensorRT provide optimization pipelines for model conversion and deployment on edge platforms. These toolchains support hardware-aware optimization and automatic conversion of full-precision models to compressed and efficient edge-compatible binaries.
- Federated learning and edge collaboration: To alleviate the need for centralized processing, Federated Learning (FL) enables distributed training across devices while preserving data privacy. This approach reduces bandwidth consumption and computational overhead on individual nodes. In addition, split learning and edge-cloud co-inference can dynamically shift processing loads between edge and cloud based on system context.

6.3.2 | Real-World Example

In a wearable healthcare IoT system developed by Zhao et al. (2021), a quantized BiLSTM model was successfully deployed on embedded wearable sensors. The optimization process reduced energy consumption by over 40% and improved battery life while maintaining an F1-score above 92%. Similarly, a smart agriculture monitoring system employed a MobileNetV2-based classifier compressed with TensorFlow Lite for anomaly detection, operating continuously on a solar-powered device with less than 300 mW consumption.

6.4 | Challenges of Federated Learning in IoT Environments

Federated learning (FL) has emerged as a promising paradigm for enabling decentralized anomaly detection in IoT networks, allowing model training across distributed devices without transferring raw data to a central server. This approach enhances data privacy and reduces transmission risks. However, despite its advantages, the deployment of FL in real-world IoT scenarios presents several critical challenges that require careful consideration and targeted research.

6.4.1 | Technical Challenges

6.4.1.1 | Communication Overhead. FL involves iterative exchanges of model parameters between edge devices and a central aggregator, which can be costly in terms of bandwidth and energy, particularly in low-power or intermittently connected IoT networks.

6.4.1.2 | Privacy Guarantees. Although raw data is not shared, model updates can leak sensitive information through gradient inversion or membership inference attacks. Differential privacy and secure aggregation are proposed solutions but introduce additional computational and statistical complexities.

6.4.1.3 | Model Heterogeneity. IoT devices vary significantly in computational capabilities and sensor modalities, leading to non-IID (non-independent and identically distributed) data distributions and divergent local models. This heterogeneity can degrade the global model's convergence and performance.

6.4.1.4 | Convergence and Scalability. The asynchronous and unbalanced nature of IoT data, combined with the unreliable participation of edge nodes, complicates global model convergence. Existing FL algorithms often struggle to scale efficiently in such dynamic environments.

6.4.2 | Emerging Solutions and Research Directions

To overcome these limitations, recent studies have proposed the following approaches:

6.4.2.1 | Adaptive Federated Optimization Algorithms. Techniques such as FedProx, Scaffold, and FedNova introduce modifications to standard federated averaging (FedAvg) to improve convergence under heterogeneous conditions. These methods incorporate proximal terms, variance reduction, and adaptive learning to stabilize updates.

6.4.2.2 | Communication-Efficient FL. Gradient compression (e.g., top-k sparsification, quantization), periodic model synchronization, and local update strategies reduce communication load while maintaining model fidelity. For example,

the SparseFed framework reduces bandwidth by over 80% with minimal accuracy loss.

6.4.2.3 | Privacy-Preserving Mechanisms. Differentially private FL and secure multiparty computation (SMC)-based aggregation protect against inference attacks, though with trade-offs in accuracy. Lightweight privacy mechanisms tailored for constrained devices remain an open research area.

6.4.2.4 | Personalized and Hierarchical FL. To handle model and data heterogeneity, personalized FL allows each device to maintain a local model tailored to its data, while hierarchical FL introduces intermediate aggregation layers (e.g., fog nodes) for scalable training across large networks.

6.4.3 | Future Research Directions

6.4.3.1 | Federated Continual Learning. Incorporating continual learning into FL to support dynamic environments where IoT data evolve over time.

6.4.3.2 | Cross-Silo and Cross-Device Hybrid FL Architectures. Combining stable institutional nodes (e.g., gateways or base stations) with transient edge devices to balance model robustness and adaptability.

6.4.3.3 | Energy-Aware Federated Scheduling. Developing federated training protocols that dynamically adjust participation based on battery levels, communication cost, and processing capability.

6.4.3.4 | Integration With Explainable AI (XAI). Integrating FL with XAI techniques to ensure interpretable local decisions and accountable global models, which are crucial in high-stakes applications such as healthcare or critical infrastructure monitoring.

6.5 | Deployment Challenges: From Lab to Real-World IoT Environments

While deep learning models for anomaly detection in IoT environments have shown promising results in academic settings, their transition to real-world deployment faces numerous practical obstacles that must be systematically addressed for sustainable and secure operationalization.

6.5.1 | Model Update and Maintenance

IoT systems are inherently dynamic, with new devices, protocols, and behaviors continually introduced. Static models quickly become outdated, necessitating continuous model updates to maintain detection accuracy. However, real-time or periodic updates in distributed environments are often limited by bandwidth constraints and computational capacity.

6.5.2 | Model Drift

The statistical properties of IoT traffic can evolve over time due to firmware updates, user behavior changes, or environmental factors, leading to model drift. If unaddressed, this results in increased false positives or missed anomalies. Drift detection techniques and adaptive learning mechanisms are essential for maintaining long-term model validity.

6.5.3 | Security of Model Transmission

When models or model updates are transferred across networks (especially in federated or distributed learning scenarios), they become vulnerable to *adversarial tampering, model poisoning, or eavesdropping*. Secure transmission protocols, encrypted model exchange, and authentication mechanisms must be employed to ensure trustworthiness and integrity.

6.5.4 | Integration With Existing Security Infrastructure

Many industrial and enterprise environments already deploy layered security tools such as firewalls, SIEM systems, or rule-based IDS. Effective deployment of AI-based anomaly detection requires *seamless integration with existing infrastructure*, ensuring compatibility with existing data formats, alert pipelines, and response workflows. This also includes aligning AI-driven insights with human-in-the-loop decision-making processes.

6.5.5 | Future Directions

- Developing online learning frameworks that support incremental updates with minimal retraining.
- Using edge-AI strategies to perform real-time inference and local adaptation directly on IoT devices.
- Designing interoperability standards for AI-driven IDS to interface with legacy cybersecurity systems.
- Exploring secure model update channels using blockchain or secure enclaves (e.g., Intel SGX).

6.6 | Model Generalization and Transferability

DL models trained on specific datasets often struggle to generalize across diverse IoT environments due to varying device behaviors, protocols, and network conditions.

6.6.1 | Proposed Solutions

- Transfer Learning enables model reuse across domains with minimal fine-tuning.

- Domain adaptation techniques align feature distributions between source and target domains.
- Case Study: A cloud-based IDS framework [40] used transfer learning to adapt a model trained on CICIDS2017 for a smart grid dataset, improving AUC by 12%.

6.7 | Security of Deep Learning Models (Adversarial Attacks)

DL models are vulnerable to adversarial inputs crafted to evade detection. Such attacks can lead to significant misclassifications, jeopardizing the security of critical IoT infrastructure.

6.7.1 | Proposed Solutions

- Adversarial training, where models are trained with perturbed inputs, enhances robustness.
- Certified defenses and ensemble learning offer further protection.
- Case Study: In an industrial control system simulation (Huang et al. 2022), integrating adversarial training into an RNN-based IDS improved resistance against FGSM attacks by over 30%.

6.8 | Need for Explainable Models

Interpretability is vital in IoT environments, especially in industrial, healthcare, and safety-critical applications. Understanding why a model flags an anomaly can support trust, compliance, and system debugging.

6.8.1 | Proposed Solutions

- Explainable AI (XAI) tools such as LIME, SHAP, and attention mechanisms provide transparent insights into model decisions.
- Hybrid models combining rule-based logic with deep learning have also gained traction.
- Case Study: A smart grid monitoring system [41] utilized SHAP values to explain transformer overload detection, enhancing stakeholder trust and decision-making speed.

6.9 | Real-Time Detection, Latency, and Throughput in Dynamic IoT Environments

In dynamic IoT environments, timely anomaly detection is critical to ensuring the security and stability of the system. However, many deep learning-based intrusion detection systems suffer from high inference latency and limited throughput, especially when deployed on constrained edge devices. These limitations hinder the ability to respond swiftly to threats and process the continuous stream of data generated by heterogeneous IoT nodes.

Achieving real-time performance requires models that can process data with minimal delay while maintaining high accuracy. Traditional batch learning approaches are often insufficient in rapidly changing environments, where new attack patterns can emerge frequently. As a result, there is a growing need for adaptive and lightweight solutions that can scale effectively with evolving traffic conditions.

Incremental learning offers a promising approach by enabling models to update continuously as new data becomes available. This eliminates the need for frequent full retraining and allows the system to adapt in real time to novel threats without significant overhead.

Model pruning, on the other hand, helps reduce the computational complexity by eliminating redundant parameters from neural networks. Pruned models retain their predictive performance while significantly lowering memory usage and inference time—making them ideal for deployment on edge or embedded IoT devices.

To further enhance system responsiveness, quantization techniques convert high-precision weights into lower-bit representations, decreasing computational load and power consumption without degrading model performance. Additionally, pipeline parallelism and edge–cloud collaborative architectures allow for balanced distribution of processing tasks between local devices and centralized servers, optimizing both latency and throughput.

Together, these approaches form a robust foundation for building high-performance anomaly detection systems that meet the real-time demands of modern IoT deployments.

7 | Challenges of Resource Consumption and Interpretability in IoT Anomaly Detection

Despite the promising accuracy of deep learning models for IoT anomaly detection, significant challenges remain related to their high resource demands and limited interpretability. These challenges are exacerbated by the inherent heterogeneity and large scale of IoT environments, where devices vary widely in computational capacity, network connectivity, and operational constraints.

7.1 | Resource Consumption Bottlenecks

- Computational complexity: Models such as Transformers and GANs often require extensive floating-point operations and memory bandwidth, which are infeasible for low-power IoT edge devices.
- Energy efficiency: Continuous anomaly detection on resource-constrained devices demands energy-efficient inference, which current large models struggle to provide.
- Latency and real-time constraints: High model complexity may lead to unacceptable delays, undermining real-time threat detection that is crucial in safety-critical applications.

7.2 | Interpretability Challenges

- Black-box nature: Deep neural networks operate as black boxes, limiting stakeholders' trust and complicating incident response.
- Lack of explainability tools: While some methods like SHAP and LIME exist, their applicability to time-series or streaming IoT data remains limited.

7.3 | Proposed Solutions and Research Directions

- Model compression and acceleration: Techniques such as pruning, quantization, and hardware-aware neural architecture search (NAS) should be prioritized to tailor models for IoT edge devices without substantial accuracy loss.
- Modular and hierarchical architectures: Designing lightweight modules that perform preliminary filtering on-device, combined with cloud-based deep analytics, can balance resource use and detection performance.
- Explainable AI (XAI) for IoT: Developing domain-specific interpretability frameworks that integrate temporal and spatial context can improve transparency and operator trust.
- Federated and collaborative learning: Leveraging distributed learning paradigms can reduce communication overhead and enable privacy-preserving model updates across heterogeneous devices.
- Adaptive and context-aware models: Future research should focus on adaptive models that dynamically adjust complexity based on resource availability and threat levels.

Addressing these bottlenecks is critical to enabling scalable, trustworthy, and efficient IoT anomaly detection systems suitable for real-world deployment.

7.4 | Explainable Ai Techniques for IoT Anomaly Detection: Current Methods and Open Gaps

As deep learning models become increasingly complex and are deployed in safety-critical IoT systems, the demand for explainable AI (XAI) grows in parallel. Explainability is essential not only for model debugging and regulatory compliance but also to build trust among human operators who rely on automated alerts for decision-making.

7.4.1 | Feature Attribution Techniques

These methods aim to identify which input features contribute most to a given prediction.

- SHAP: Based on cooperative game theory, SHAP assigns importance scores to features for each prediction. SHAP is model-agnostic and supports both tabular and time-series data.
  - *IoT Application*: SHAP has been successfully used in anomaly detection for smart grid monitoring and wearable healthcare systems to highlight which features (e.g., voltage drop, CPU usage) signal anomalies.
  - *Limitation*: *Computationally* expensive for large models or streaming data; interpretation becomes less intuitive for high-dimensional feature spaces.
- LIME: Generates local surrogate models to approximate the behavior of black-box models around a specific instance.
  - IoT *Application*: Effective in simple sensor environments with static feature spaces.
  - Limitation: *Struggles* with sequential dependencies in time series data, which are common in IoT networks.

7.4.2 | Attention Mechanisms

Originally designed for natural language processing, attention mechanisms have been widely adopted in deep learning for IoT anomaly detection—especially in recurrent architectures like LSTM and Transformer models.

- *Use Case*: Attention allows models to focus on important time steps or feature dimensions. For example, in industrial control systems, attention weights can reveal which sensor readings or time intervals were critical in identifying a malfunction.
- *Strength*: Naturally provides interpretability in temporal and multivariate contexts.
- *Limitation*: Attention does not always correlate with true feature importance and may mislead when improperly visualized or aggregated.

7.4.3 | Prototype Learning and Concept Bottlenecks

These emerging techniques seek to *map anomalies to human-interpretable prototypes or concepts*, rather than relying purely on numerical feature importance.

- *Example*: Concept Bottleneck Models (CBMs) introduce intermediate interpretable concepts (e.g., "high temperature anomaly") before the final decision layer.
- *IoT Potential*: Useful in smart homes, autonomous vehicles, and medical IoT where domain-specific intermediate concepts can be explicitly defined.
- *Limitation*: Requires manual annotation or domain knowledge to construct concept sets.

7.4.4 | Visualization-Based Techniques

- *Time-Series Saliency Maps*: Highlight critical time intervals or channels responsible for predictions.
- *Grad-CAM Adaptation for 1D-CNNs*: Enables visual attribution in 1D sensor data streams by showing which segments influence output classes most.

7.4.5 | Hybrid Symbolic-Neural Models

Combining deep learning with rule-based systems or decision trees can enhance interpretability:

- *Example*: Use CNNs to extract features and symbolic logic to explain anomalies (e.g., "anomaly due to repeated TCP SYN packets").
- *Advantage*: Retains accuracy of deep models while offering structured explanations.

8 | Discussion

Despite the remarkable success of deep learning models in anomaly detection for IoT networks, several critical challenges remain unaddressed, particularly regarding their robustness against adversarial attacks. In real-world applications, attackers can exploit model vulnerabilities by introducing carefully crafted perturbations—known as evasion or adversarial attacks—to deceive detection systems without altering the semantic characteristics of the malicious behavior. This poses a significant threat, especially in safety-critical IoT environments such as healthcare, industrial control systems, and smart grids.

Recent studies have demonstrated that deep neural networks, including CNNs, LSTMs, and even GAN-based detectors, can be sensitive to minor input alterations. Therefore, enhancing model robustness has become a crucial aspect of ongoing research in secure AI. Adversarial training, in which models are trained with both clean and adversarially perturbed data, has proven effective in improving resilience. Additionally, defense mechanisms such as feature squeezing, input denoising, model ensembling, and defensive distillation offer promising pathways to reduce susceptibility to evasion.

Another emerging solution involves incorporating generative models like GANs to simulate adversarial samples during training, enabling detectors to recognize a broader range of deceptive patterns. Moreover, hybrid architectures that combine statistical analysis with deep learning have shown increased robustness by leveraging diverse decision boundaries.

To promote the safe deployment of IoT anomaly detection systems, it is essential to integrate these defensive strategies into existing models. Furthermore, standardized adversarial benchmarks and robustness evaluation protocols should be established to fairly assess and compare the resilience of different approaches under adversarial conditions.

In future work, we recommend expanding evaluations to include adversarial scenarios and testing models against various attack vectors such as FGSM, PGD, and Carlini-Wagner. Ultimately, achieving robust, reliable, and adaptive anomaly detection systems in IoT requires a balanced combination of accuracy, generalization, and security-aware model design.

The integration of traditional statistical techniques with deep learning models represents a powerful and emerging direction for enhancing anomaly detection performance in IoT networks. Statistical methods—such as time-series decomposition, entropy analysis, and change-point detection—offer interpretability, computational efficiency, and domain-driven insights that are often lacking in complex neural models. When combined with the adaptability and representation power of deep learning, particularly recurrent or generative architectures, these hybrid approaches can capture both macro-level patterns and subtle temporal anomalies. For instance, statistical preprocessing can enhance data quality and highlight informative features before feeding them into a deep model, while post-processing statistical validation can reduce false positives by cross-verifying predictions. This complementary synergy not only improves detection accuracy and robustness but also enhances model explainability, which is critical for deployment in sensitive domains such as industrial control systems, healthcare, and energy infrastructures. Emphasizing this interaction paves the way for the development of more practical, interpretable, and trustworthy AI-based detection frameworks in real-world IoT environments.

Moreover, recent advancements in deep learning architectures, particularly Graph Neural Networks (GNNs) and Transformers, have demonstrated promising capabilities in capturing complex spatial–temporal dependencies and global contextual information. GNNs enable anomaly detection by modeling communication patterns and relationships between IoT nodes as structured graphs, while Transformers facilitate robust temporal modeling in sequential IoT data. Incorporating these models into IoT anomaly detection pipelines may further enhance accuracy, interpretability, and generalization across heterogeneous IoT environments. Future research should consider leveraging such architectures, especially in combination with existing defense mechanisms, to develop next-generation secure and intelligent intrusion detection systems.

8.1 | Discussion on Hybrid Architectures

While traditional deep learning models such as CNNs, LSTMs, and Transformers have been extensively studied, hybrid architectures remain relatively underexplored in IoT anomaly detection. Hybrid models can integrate rule-based systems, statistical methods, or ensemble techniques with neural networks to leverage the complementary strengths of each approach. For example, combining CNN-LSTM networks with lightweight rule-based filters can improve the detection of rare anomalies while reducing false positives. Ensemble methods, such as averaging or stacking predictions from multiple models, can enhance robustness to noise and adversarial perturbations. However, hybrid approaches may introduce additional computational overhead and system complexity, requiring careful design to balance accuracy, latency, and resource consumption. Including preliminary experiments or case studies on hybrid models can provide valuable insights into their practical advantages and limitations, guiding future research in designing more resilient and efficient IoT anomaly detection systems.

9 | Future Directions and Research Trends

Recent advances in deep learning for anomaly detection in IoT networks have opened numerous avenues for further research. To

bridge the gap between academic progress and practical deployment, future efforts should emphasize actionable and impactful directions, including but not limited to:

1. Development of Explainable AI (XAI) Models Tailored for IoT
   There is an increasing demand for transparent and interpretable models that can provide clear rationales for anomaly detection decisions. Explainable AI techniques adapted to the specific constraints and data characteristics of IoT environments will enhance trust and facilitate compliance with security policies and regulations.
2. Design of Adaptive and Context-Aware Detection Methods
   Given the rapidly evolving nature of cyber threats and IoT network behaviors, future models should incorporate adaptive mechanisms capable of detecting novel or stealthy attacks. Techniques such as online learning, domain adaptation, and context-aware anomaly detection can enable models to dynamically adjust to changing environments without full retraining.
3. Implementation of Continuous and Lifelong Learning Strategies
   IoT networks are inherently dynamic, with evolving device populations and usage patterns. Continuous learning frameworks that update models incrementally using streaming data can maintain detection performance over time, mitigate concept drift, and reduce the need for costly retraining on large static datasets.
4. Lightweight and Energy-Efficient Architectures
   Considering the limited computational and power resources of many IoT devices, developing efficient deep learning models that balance accuracy with resource constraints remains critical. Model compression, pruning, quantization, and edge computing paradigms are promising approaches.
5. Federated and Privacy-Preserving Learning
   To address privacy concerns and bandwidth limitations in IoT, federated learning approaches that train models across distributed devices without sharing raw data are essential. Future research should explore scalable and robust federated methods suited for heterogeneous IoT systems.
6. Creation of Realistic, Multimodal, and Multidimensional Datasets
   The availability of diverse datasets representing real-world IoT scenarios, including multimodal sensor data and complex attack vectors, is vital for advancing model generalizability and robustness. Collaborative efforts to develop and share such datasets will accelerate research progress.

While recent research highlights the importance of model robustness, empirical evaluation against adversarial attacks remains limited. Future studies should systematically incorporate adversarial sample generation (e.g., FGSM, PGD, Carlini-Wagner) and assess the effectiveness of defense mechanisms such as adversarial training, feature squeezing, input denoising, and model ensembling. By integrating targeted experiments on adversarial scenarios, researchers can better validate model stability, enhance resilience in real-world IoT deployments, and provide stronger guarantees for safety-critical applications.

In summary, focusing on explainability, adaptability, and continuous learning in the context of resource-aware architectures will be key to realizing robust and deployable anomaly detection solutions in IoT networks.

9.1 | Enhancing Replicability Through Detailed Experimental Descriptions

Although the paper references case studies in domains such as industrial IoT and smart homes, the lack of detailed experimental setups limits replicability and practical relevance. Future work should provide comprehensive documentation of data preprocessing steps, model architectures and hyperparameters, training protocols, and deployment environments. Including such details enables other researchers to reproduce results, compare methodologies consistently, and better understand the operational constraints and challenges faced in real-world IoT deployments. This transparency not only strengthens scientific rigor but also bridges the gap between academic research and practical applications.

9.2 | Quantitative Trade-Offs: Accuracy Versus. Resource Constraints in Lightweight Deep Learning Models

The integration of deep learning in IoT-based anomaly detection must account for the fundamental limitations of edge and embedded devices, including restricted computational power, memory, and energy capacity. Although many studies have proposed accurate detection models, few have systematically benchmarked these models in terms of resource usage — an essential requirement for real-world deployment in constrained IoT environments.

Several recent studies have investigated the balance between detection performance and model efficiency. Table 9 summarizes a comparative analysis based on results drawn from empirical research and simulation environments, including works such as [1, 4, 8, 10, 14–16, 26, 30, 31, 33, 42].

9.3 | Key Observations and Trade-Offs

1. Accuracy versus efficiency: While LSTM and hybrid CNN-LSTM architectures achieve high accuracy (˜97%–98%), they incur high computational overhead, making them less suitable for edge deployment in battery-powered devices.
2. Lightweight models:
   - *MobileNetV2* and *SqueezeNet* show strong performance with over 93% accuracy, while drastically reducing inference latency and memory footprint — making them ideal for real-time IoT applications.
   - Quantized *and* pruned models retain comparable accuracy to full-size models while reducing memory by 60%–70%.

TABLE 9 | Benchmark comparison of DL models for IoT anomaly detection.

Model	Accuracy (%)	Model size (MB)	Inference time (ms)	Energy consumption (mW)	References
CNN (Baseline)	96.2	12.5	230	~520	[1, 33]
LSTM	97.4	17.8	280	~590	[4, 30, 42]
MobileNetV2	94.6	3.4	78	~270	[8, 14, 26]
SqueezeNet	93.1	2.8	64	~250	[8, 14]
Quantized LSTM	96.1	4.7	98	~310	[8, 15–16]
Pruned BiLSTM	95.3	5.1	105	~340	[15, 31]
Autoencoder (small)	91.8	2.3	52	~240	[4, 33]
Kolmogorov–Arnold Net	92.5	5.2	101	~335	[25, 27]

3. Generative and hybrid models:
   - Lightweight GANs and KANs (Kolmogorov–Arnold Networks) show promising results in noisy environments ([15, 25]) but require tuning to remain stable under real-time constraints.
4. Energy-aware deployment: Energy usage below 300 mW is essential for deployment on platforms like Raspberry Pi Zero, ESP32, or ARM Cortex-M. Models such as SqueezeNet and compressed autoencoders meet this criterion.

9.4 | Best Practices for Deployment on IoT Devices

- Use compressed CNN variants (e.g., MobileNetV2, SqueezeNet) for general-purpose, fast inference.
- Adopt quantization-aware training to minimize degradation during model conversion.
- Implement pruning and knowledge distillation when deploying LSTM or Transformer-based models.
- Edge-AI toolkits (e.g., TensorFlow Lite, ONNX Runtime, PyTorch Mobile) should be integrated with monitoring for real-time anomaly detection.

As highlighted in [8, 10, 14, 26], the future of anomaly detection in IoT lies in the synergy of performance and efficiency. Selecting models based solely on accuracy is inadequate for deployment in constrained environments. Future research must include standardized evaluation metrics that measure power consumption, memory usage, and processing latency—ensuring realistic, scalable, and sustainable deployment across diverse IoT settings.

10 | Conclusions

This review paper provided a comprehensive analysis of deep learning techniques for anomaly detection in IoT networks, highlighting their advantages and limitations. Various models, including CNN, LSTM, Autoencoders, GANs, and hybrid architectures, were compared in terms of accuracy, training and inference time, and resource consumption.

Key challenges such as noisy and imbalanced data, the complexity of real-time processing, limited device resources, and the need for interpretable models were discussed in detail. The paper emphasized the importance of developing lightweight models, leveraging transfer learning, integrating statistical methods with deep learning, and adopting federated learning for improved performance and privacy.

Moreover, advancements in creating realistic, multidimensional datasets and designing self-explaining models will significantly contribute to enhancing the effectiveness and reliability of anomaly detection systems in dynamic and complex IoT environments. Continued research in these directions is essential for building robust, efficient, and trustworthy IoT security frameworks.

Author Contributions

Roya Morshedi: conceptualization, investigation, writing – original draft, writing – review and editing, methodology, formal analysis, resources, data curation, software, validation, project administration. **S. Mojtaba Matinkhah:** supervision.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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